A Computer Vision Tool to Aid Beginner Electric Bass Players with Fretboard Navigation

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Abstract

This paper presents early research into the development of a computer vision-based learning tool designed to assist novice electric bass guitarists. The fretboard, strings, hand and finger positions are automatically detected. Musical input is parsed from standard bass tablature and synchronised with real-time finger placement and plucking detection to generate visual cues on the live video feed of their guitar for correct note positioning.

Keywords: Computer Vision Application, Edge Detection, Hand Pose Estimation, Optical Flow Estimation.

1 Introduction

For beginners learning to play the electric bass guitar, one of the most significant early challenges is navigating the fretboard [Harrison, 2010]. Traditional methods, such as printed tablature, require users to mentally translate notation into physical movements on the instrument, often leading to cognitive overload, slow progress, and discouragement. This paper presents a novel computer vision-based learning tool designed to assist novice bassists by over laying real-time visual guidance directly onto their instrument. A live video feed of the player's bass guitar, detects the fretboard and the four strings (typically tuned to E, A, D, and G), and tracks hand and finger positions using the MediaPipe Hands framework. Musical input is parsed from bass tablature and synchronised with real-time finger placement and plucking detection to generate on-instrument visual cues for correct note positioning. The system enforces deliberate practice by advancing only when the correct note is detected (See Figure 1).

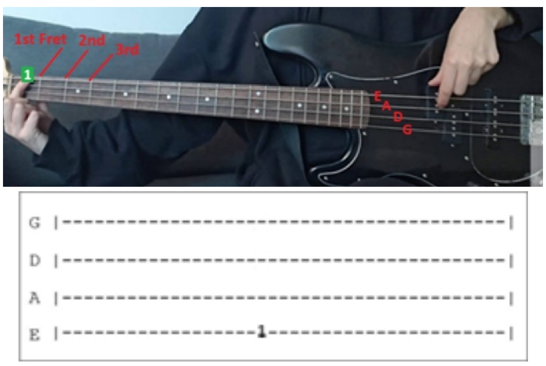

Figure 1: The relationship between the fretboard and the tablature (bottom), with the E string to be plucked while the 1st fret is held down

2 State of the Art

Much work in this field has concentrated on Automatic Music Transcription (AMT) as opposed to aiding learning. Video is used in AMT as audio is insufficient (as the same note can often be played on multiple strings and at various fret positions in string instruments). This redundancy creates ambiguity in note identification when using audio-only systems. This limitation is particularly problematic for generating tablature notation, which is essential for guitarists and bassists who rely on precise finger placement information for learning and performance. For example, [Asmar, 2022] presented a novel approach for the automatic transcription of guitar music using computer vision and depth data from RGBD cameras. The process for detection begins with fretboard segmentation, where a deep learning model is trained to produce a binary mask of the fretboard area. This segmentation isolates the fretboard from the background, improving the accuracy of subsequent steps such as string and fret detection. Additionally, the system uses optical flow analysis to detect string plucking or picking motions, which is essential because pressing a string without plucking does not produce sound. Their system focuses on generating tablature by solely relying on visual input, effectively eliminating the complexities associated with audio signal processing.

[Ghaleb et al., 2024] present a system that generates guitar tablature by combining video inputs with audio analysis. The visual component uses deep learning models to detect the fretboard and hand positions, even in challenging conditions such as occlusions or rapid note transitions. On the audio side, the system applies the Fourier Transform to convert time-domain signals into frequency- domain data, extracting precise note information.

The most similar work we could locate was by [Elashmawi et al., 2023] who present a hybrid system that integrates finger motion capture with note frequency recognition to assess guitarists' performance. This system does offline analyses videos being played by guitarists using acoustic guitars.

3 System Design

Our system has two inputs: (1) a live camera feed from a stationary camera showing the user playing the guitar and (2) the tablature of the music to be played. The live camera feed is processed to find the fretboard and the hands, and the tablature is processed to track the current/next note to be played.

Visual guidance is overlaid on the video feed to show finger placement and to highlight the note on the tablature. See Figure 2.

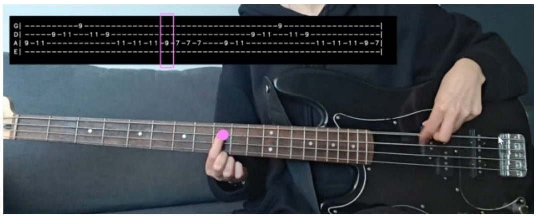

Figure 2: Live video feed with a highlight (pink dot) indicating where the player should be depressing the string/fret (the 9th fret on the A string). The tablature which is being played is shown above with the relevant note highlighted in the pink rectangle

3.1 Bass guitar recognition

The fretboard is located using edge detection and the Hough transformation. The dominant near horizontal lines will come from the guitar strings. These are used to rotate the image so the guitar strings are roughly horizontal. The frets will then be roughly orthogonal lines. The positions of the frets relative locations are defined by the *12th-root-of-2* rule. See Figure 3. This formula defines the distance from the nut to the n-th fret as

$$x_n = \mathrm{L} \times \left(1 - 2^{-n/12}\right) \quad (1)$$

where x_n is the distance from the nut to the n-th fret, L is the total scale length (distance between the nut and the bridge), and n is the fret number.

Figure 3: Fret locations determined from the from the vertical lines between the strings using the *12th-root-of-2 rule*

3.2 Plucking detection

MediaPipe Hands (Zhang, 2020) is used to detect the hands and finger positions of each hand. The hands are classified as being the fretting hand or the plucking hand depending on their position relative to the fretboard.

The fingertip positions are used both to determine what string is plucked and what fret of that string is depressed. Plucking is detected using a combination of fingertip tracking and motion analysis. The index and middle fingers of the plucking hand are tracked and the distance from each of the strings determined. A potential pluck is detected as a fingertip remaining near one string for several frames followed by a rapid upward transition to another string (or roughly that distance). Optical flow is used on the potentially plucked string to verify that motion has occurred on that string. Once a pluck is verified by movement on the string, the fret position is determined by analysing the fingertip

Figure 4: A pluck being detected by the system

position of any finger overlapping the plucked string. See Figure 4.

3.3 Musical Processing

The system accepts musical input in the form of tablature (a format which is widely used and intuitively understood by beginner bass players). The tablature is read from left to right and shows 4 lines (one for each bass guitar string G, D, A, E). Fret numbers are indicated on one string at a time indicating what string should be plucked and the number indicates what fret should be depressed. Fingertip positions are used both to determine what string is plucked and what fret of that string is depressed. See Figure 5.

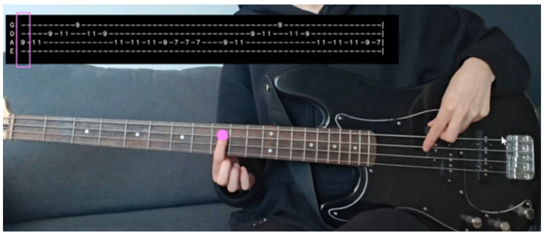

Figure 5: The system interface showing the tablature (top) and a live video feed of the guitar with a pink dot highlighting the string/fret to be played

Once a tablature is loaded the system displays the tablature over a live view of the bass guitar being played. The note to be played is highlighted in the tablature (by putting a pink rectangle around it) and the string and fret position are highlighted on the live video of the guitar (by putting a solid pink circle in the location of the fret which should be depressed.

Only once the relevant fret is depressed and the relevant string plucked does the system move onto the next note on the tablature.

4 Evaluation

The system was tested by (1) objectively testing how accurately notes were recognised, and (2) subjectively assessing how useful the system seemed to users.

4.1 Assessment of note recognition

To assess the performance of the Fret-Nav system to recognise the correct notes played, a test video was recorded in which a sequence of 157 individual notes were plucked across all four strings of a standard four string precision model electric bass guitar. The notes detected were compared manually with the notes actually played resulting in an accuracy of 91% and a precision of 94% (See Table 1).

	Actual Positive	Actual Negative
Predicted Positive	151	9
Predicted Negative	6	-

Table 1: Pluck detection summary.

Most false positives were caused by ambiguous plucking gestures and finger movement in the test video that were misclassified as note events. A significant amount of the false positives (≈ 44%) were concentrated on the E string. This can likely be attributed to the string's position at the top of the instrument and the lack of a higher string above it, a factor that limits the system's ability to verify plucking gestures through motion patterns that depend on placing a finger on the string above the one being played, as it does for the A, D, and G strings. The increased sensitivity of the E string detection is a side effect of the model's design and although a phantom string is used to account for the differences of the top string, it may require future calibration or separate plucking detection rules to prevent over-triggering.

False negatives were most commonly caused by errors in finger tracking. In several instances, a string pluck was correctly identified, but the MediaPipe-based finger tracking component reported fingertip positions that fell just outside the expected fret region. This could be due to imprecise fingertip landmark estimation or noise in the video data. These results highlight that the reliability of note identification is highly dependent on the accuracy of both components — pluck detection and fret positioning — and even minor inaccuracies in either can result in missed detections.

4.2 User study

Our initial subjective assessment of the system involved six different participants with varying levels of prior musical experience but no experience of playing the electric bass guitar. All participants received a short introduction to two-finger plucking technique and how to read tablature. They were given one hour to learn and practice the same sequence. Half of the participants used the traditional method to learn and the other half used the Fret-Nav system.

Following the session, each participant completed a structured survey combining Likert Scale questions and open-ended responses. Table 2 provides an overview of the study results for the Likert Scale questions.

Survey Category	Traditional Group	Fret-Nav Group
Engagement	Agree to Strongly Agree	Agree to Strongly Agree
Ease of Learning	Neutral to Agree	Agree to Strongly Agree
Learning Method Effectiveness	Mostly Agree	Mostly Strongly Agree
Need for Additional Guidance	Neutral to Agree	Neutral to Disagree
Ability to Self-Correct	Mostly Neutral	Mostly Agree

Table 2: Summary of user survey results.

Subjectively, the system improved learners' confidence, self-correction ability, and engagement compared to traditional learning methods. Participants viewed the tool as intuitive, effective, and enjoyable. While areas for refinement were identified — such as pluck detection calibration and UI adjustments — the results from this pilot study strongly support the tool's value as a beginner-oriented learning aid. The combination of real-time feedback and intuitive visual guidance clearly reduces the cognitive load associated with early-stage fretboard navigation.

5 Conclusions

This paper presents a novel live application of computer vision in music education for the bass guitar. A prototype system was developed and tested showing potential to aid in the learning process. The initial system described in this paper achieved a note detection accuracy of 91% and precision of 94%. A user study comparing the use of Fret-Nav to traditional learning techniques showed increased engagement, self-correction ability, and reduced cognitive load among participants using the tool.

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